Part 3 - Manual Gradient Descent : Iteration 2

Relay input (from Arnold's Iteration 1):

$m_{new} = [-1.45, 0.87]$ ~~[...]~~

$b_{new} = [0.99, 0.89]$

Data

$X = [(1,3), (4,10)]$

$Y = [5,6), \alpha = 0.01, n = 2]$

Predict $\hat{Y} = X \cdot m + b$

$X \cdot m$:

First row : $(1)(-1.45) + (3)(0.87) = -1.45 + 2.61 = 1.16$

Second row : $(4)(-1.45) + (10)(0.87) = -5.80 + 8.70 = 2.90$

$X \cdot m = [1.16, 2.90]$

$\hat{y} = X \cdot m + b = [1.16 + 0.99, 2.90 + 0.89] = [2.15, 3.79]$

Error $e = \hat{y} - y$

$e = [2.15 - 5, 3.79 - 6] = [-2.85, -2.21]$

Cost J (MSE)

$J = (1/n) \cdot \sum e_i^2$

$J = \frac{1}{2}[(-2.85)^2 + (-2.21)^2]$

$= \frac{1}{2}(8.1225 + 4.8841)$

$= 1/2 (13.0066)$

$J = 6.5033$

Gradient of J with respect to m

$\partial J/\partial m = (2/n) X^T e$

$X^T = [1,4], [3,10]$

$X^T e$:

row 1 : $(1)(-2.85) + 4(-2.21)$

$= -2.85 - 8.84 = -11.69$

row 2 $= (3)(-2.85) + 10(-2.21)$

$= -8.55 - 22.10 = -30.65$

$\partial J/\partial m = (2/2) \times [-11.69, -30.65]$

$= -11.69, -30.65$

Gradient of J with respect to b

$\partial J/\partial b = (2/n)\, e_i$

$= 2/2 \times [-2.85, -2.21]$

$= -2.85, -2.21$

new value = old value - (α x gradient)

$M_2 = M_1 - \alpha \cdot \partial J/\partial m$

$= [-1.45, 0.87] - 0.01 \times [-11.69, -30.65]$

$= [-1.45 + 0.1169, 0.87 + 0.3065]$

$= [-1.3331, 1.1765]$

$b_2 = b_1 - \alpha \cdot \partial J/\partial b$

$= [0.99, 0.89] - 0.01 \times [-2.85, -2.21]$

$= [0.99, 0.89] - [-0.0285, -0.0221]$

$[0.99 + 0.0285, 0.89 + 0.0221]$

$[1.0185, 0.9121]$

$\therefore M_2 = [-1.3331, 1.1765]$

~~$b_2 = [1.0$~~

$b_2 = [1.0185, 0.9121]$